MAJOR PROJECT SYNOPSIS · B.TECH COMPUTER SCIENCE & ENGINEERING · PROJECT TEAM ID CSE27-293

MFSTNet — Multimodal Spatio-Temporal Congestion Forecasting for Adaptive Traffic Signal Control

Forecast per-lane congestion 60 seconds ahead from an existing traffic camera alone, and let a reinforcement-learning controller act on it — no new roadside sensor, no raw frame leaving the camera, and a safe fallback when either the forecast or the network is lost.

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What this synopsis proposes

The deck follows the synopsis chapter order, so a panel reading along can follow.

- **1 Introduction**
  Background, motivation, and why a camera-only route is the one worth taking
- **2 Literature Review**
  Four research strands, each mature alone, and the gap between them
- **3 Problem Statement & Objectives**
  One precise problem; six objectives O1-O6 that discharge it
- **4 Proposed Methodology**
  Dataset, detector, MFSTNet, PPO controller, prototype, evaluation
- **5 Hardware & Software**
  What the project actually runs on — one laptop, one 6 GB GPU
- **6 Expected Outcomes**
  Seven deliverables, each with a measurable target stated in advance
- **7 Conclusion**
  Where the work stands, including the half that is not yet working
- **References**
  IEEE style, 16 entries, cited at first use in the report

One thing to hold onto throughout: the control half of this project already works and is measured; the vision half is measured and does not yet beat a count-only baseline. Both are in the report.

A signal allocates green time from a survey taken once

Fixed-time control computes cycle length and phase splits from a manual count, then holds them for years.

- **Fixed-time plans go stale**
  Webster's 1958 method is still what is deployed, and still the honest baseline — but the demand that justified a plan changes long before the plan does.

- **Adaptive control needs measurement**
  and measurement is the expensive part: loops are cut into the carriageway and fail with the surface; radar and lidar are priced per approach.

- **Both assume lanes**
  A stream of filtering two-wheelers registers as a small count while consuming the discharge capacity of a car.

WHY A CAMERA

- **Already installed**
  at a large share of urban junctions
- **Sees the whole approach**
  not a point under the tarmac
- **Runs on commodity hardware**
  detection near the camera
- **Costs nothing new**
  no civil work, no per-approach unit

What a camera does not give directly is the quantity a controller needs: not the present state of the intersection, but its state a minute from now.

Three gaps that make this worth building

Each one changes a design decision later in the deck, so they are stated first.

1

A reactive controller is always one decision behind

By the time a queue is visible it is already costing waiting time. A 60-second forecast lets a phase decision be right when it takes effect, not when it is made.

2

General detectors cannot name Indian road users

COCO has no auto-rickshaw, no e-rickshaw and no cattle class — and a detector that cannot name a vehicle cannot count it. Eight India-relevant classes are trained instead.

3

Fusion papers omit the floor a frozen-backbone study owes its reader

Without a linear probe on the same frozen features, an architectural gain and a feature-quality gain are indistinguishable. So the probe is configuration H, not an appendix.

Consequence for this project: the ablation, not the architecture, is the deliverable of scientific interest.

Four strands, each mature on its own

Every row below is well developed in isolation. The combination is what is missing.

STRAND	REPRESENTATIVE WORK	STRENGTH	LIMITATION FOR THIS PROBLEM
Detection	YOLO family [5]; IDD [6]	Real-time on modest hardware	Dashcam viewpoint dominates; CCTV view seldom reported
Forecasting	LSTM [14]; XGBoost [15]	Strong on sensor time series	Assumes an instrumented network; count-only control omitted
Fusion	ViT [2]; DINOv2 [3]; attention [1]	Complementary local and global features	Linear-probe floor rarely reported, so gains unattributable
RL control	PPO [7]; SB3 [8]; SUMO [9]	Mature algorithms and simulators	Weak or unswept baselines [10]-[12]; variance unreported

THE PATTERN

Detection work rarely carries through to control. Forecasting work rarely reports the count-only control condition. Fusion work rarely reports the linear-probe floor. Control work rarely sweeps its baseline. Each omission is individually forgivable and collectively the reason a reader cannot tell which component earned the result.

What we claim, and what we do not

The fusion mechanisms are published. Claiming them would not survive review, so the claim is narrower and defensible.

G1

Camera-only, forecast-driven control

An end-to-end path from one existing camera to a signal decision, with no additional roadside sensing, is not established for unstructured traffic.

G2

The gate reported as an artefact

Gated fusion is published as a mechanism; the gate value is treated as an internal activation. Logging it, tracking it live and analysing when it saturates turns it into evidence about which modality the model is using.

G3

Density-stratified, leave-one-camera-out evaluation

Random splits over overlapping windows leak, and a camera-hashed split can put an entire traffic regime in one partition. One held-out camera at a time, reported as a fold mean with an interval, is what makes a cross-camera claim mean anything.

Stated the other way: this is not a new fusion mechanism. It is a camera-only path to control, an interpretable gate, and an evaluation protocol strict enough that a negative result is publishable.

One sentence, four difficulties

Given only the video stream of a camera already installed at a signalised urban intersection carrying heterogeneous, weakly lane-disciplined traffic, predict the congestion level of each of the four approaches 60 seconds ahead, and use that prediction to select signal phases so that mean vehicle waiting time is lower than under the best-configured Webster plan — without installing any additional sensor, without transmitting any raw frame off the capture device, and without failing unsafely when either the forecast or the network is lost.

Counting is not detection

A detector strong on mAP can still mis-count under occlusion. The operating point is chosen to minimise counting error, not to maximise mAP.

Labelling is not free

Congestion has no ground-truth sensor here, so labels are derived from counts under a stated, per-camera-calibrated rule.

Forecasting is not description

The model must predict a state it cannot see — the label is 60 seconds past the last observed frame.

Evaluation is easily circular

Labels come from detector counts, and three baselines also consume detector counts. So the test split is human-verified.

Six objectives, each independently checkable

Every objective names an artefact that either exists or does not, so progress is not a matter of opinion.

01 Licence-audited corpus of unstructured-traffic footage; YOLOv8s fine-tuned over eight India-relevant vehicle classes.

02 Per-lane congestion labels from PCU-weighted occupancy with per-camera thresholds, so a label means the same thing on a six-lane arterial and a two-lane approach.

03 MFSTNet: gated bidirectional cross-attention over frozen CNN and ViT features, then recurrent and attention-based temporal modelling, predicting 60 s ahead.

04 Eight-configuration ablation including a linear probe, evaluated leave-one-camera-out so no reported score is a memorised camera.

05 PPO controller on a SUMO intersection whose observation carries the forecast, measured against a swept Webster baseline over 30 paired seeds with a pre-registered disqualification rule.

06 Edge-to-server prototype in which no raw frame leaves the capture device, and in which both required fallback paths are exercised.

O1-O4 are the vision and evaluation track; O5-O6 are the control and prototype track. They were built to be independent on purpose, so a delay in one does not stall the other — and O5 is already delivering a result while O4 is still gathering power.

Three tiers, one message contract

Frames never leave the edge tier. Only derived counts and predictions cross the network.

EDGE TIER team laptop, at the camera	SERVER TIER FastAPI + ONNX + PPO	PRESENTATION TIER React dashboard
Fixed CCTV camera	MFSTNet (ONNX) — 60 s ahead	Live panel — counts, predictions, gate
YOLOv8s detection — 8 classes	PPO policy — 16-dim, 12 actions	Benchmark panel — committed CSV
Per-lane counting	SignalGuard — min/max green, all-red	The two panels never merge
Local Webster fallback	SQLite — bounded history	Refresh within 2 s

```
MQTT CONTRACT    emergency QoS 2  |  counts and commands QoS 1  |  predictions and heartbeat QoS 0    —    QoS is attached to the topic, never passed by the caller
```

The coupling that is easy to miss: MFSTNet's output is not only displayed, it is part of the controller's observation, at indices 11-14 of a 16-dimensional vector. If the forecast is unavailable those indices are zeroed — the vector is never shortened, because that would silently change the meaning of every later index.

The corpus is built so a threshold is a config change

Counts are stored separately from labels, so relabelling takes seconds instead of re-running the detector over hours of video.

Source clips	curated public footage + campus capture
Triage	reject clips too short or too degraded
Lane geometry	placed by hand, then reviewed
Detection	fine-tuned YOLOv8s → per-lane counts
Counts file	threshold-independent
Labelling	PCU occupancy, per-camera p33/p67
Audit	fails loudly; nothing proceeds past it
Feature cache	frozen backbones once, fp16 per clip
Split	leave-one-camera-out, cut by source clip

DECISIONS THE CORPUS MAKES

- **Twenty two-wheelers ≠ twenty buses**
  `same count, different road`
- **Indo-HCM equivalence values**
  `car 1.00 · motorcycle 0.24 · auto 0.78`
- **Thresholds per camera**
  `p33 / p67 of that camera's own distribution`
- **Split by source clip, never by window**
  `neighbouring windows share most frames`
- **Lane centres by hand, always reviewed**
  `auto centres moved 0.164 of frame width`

An unmapped vehicle class raises rather than defaulting to a weight of 1.0.

A cache is keyed on backbone, resize and normalisation; a mismatch raises rather than warns, because a stale cache trains normally and reports wrong numbers.

MFSTNet — frozen backbones, learned fusion

Both backbones are frozen and cached, which is what makes an 8-configuration, 4-fold, 5-seed sweep fit on one 6 GB laptop GPU.

Frame sequence — T = 60 frames at 5 s

ResNet-50 (frozen) 2048×7×7 || DINOv2 ViT-S/14 (frozen) 384 tokens, CLS dropped

Aligned onto a shared 7 × 7 grid — the gate is elementwise, so the grids must match

Z_A = CNN queries ViT || Z_B = ViT queries CNN

Gated fusion — the gate value is logged, tracked and reported

BiLSTM 2 layers × 128, bidirectional → temporal self-attention 2 layers × 4 heads

Attention pooling over time + per-lane ROI pooling

Per-lane head — LOW / MEDIUM / HIGH for N, S, E, W at t + 60 s

THE GATE (4.1) AND (4.2)

$$g = \sigma(W [Z_A ; Z_B])$$

$$F = g \cdot Z_A + (1-g) \cdot Z_B$$

Retained, not discarded. A gate at 0.5 with negligible variance means averaging, not selection.

MEASURED, CONFIG G

$g = 0.5299$ $\sigma = 0.0090$

ABLATION-ABILITY IS ARCHITECTURAL

- **Every module disableable by flag**
  no code edit A → H
- **H is the linear probe**
  the floor, not an appendix

The controller, and a baseline worth beating

A weak baseline is the commonest way an RL traffic result stops meaning anything, so the baseline is swept rather than chosen.

PPO AGENT

- **Observation**
  16-dim; forecast at indices 11-14
- **Action space**
  12 discrete: {NS, EW} × {10,20,30,45,60,90} s
- **Algorithm**
  lr 3e-4 · n_steps 2048 · batch 64 · γ 0.99
- **Budget**
  clip 0.2 · ent 0.01 · 500,000 timesteps

WEBSTER BASELINE

- **Saturation flow is not chosen**
  it is swept: 525 → 1283 PCU/h/m
- **Best configuration wins**
  and that is what PPO must beat
- **Clamping is logged**
  clamp rate reported beside the result
- **Disqualification is pre-registered**
  <85% trips, or >50% clamped

REWARD (4.3)

$$r_t = -(W_t - W_{t-1}) / 100 - 1.0 \cdot n_{starved} - 0.05 \cdot 1[switch]$$

Reduction in total lane waiting time between successive decisions, penalised for starving an approach and lightly penalised for switching. Minimum and maximum green of 10 s and 90 s and a 3 s all-red are enforced by actuation, not learned — a safety constraint the agent could trade against reward is not a constraint. A Webster controller that clamps every cycle has stopped computing anything and is a fixed-time plan under another name, which is why the clamp rate is reported beside the result.

Evaluation is the deliverable

Leave-one-camera-out, five seeds, bootstrap intervals, and raw per-run CSVs committed beside the code.

- **Leave-one-camera-out**
  Each fold holds out one camera entirely — 4 folds today. A reported score cannot be the model recognising a camera it has already seen.

- **Five seeds, and the spread is reported**
  On this corpus one configuration moved by more than 0.24 macro-F1 across seeds on a single fold — larger than the gap between most pairs of configurations. The harness warns when the spread exceeds the closest gap, because those two are then not separated.

- **Ordinal metrics, not just accuracy**
  Macro-F1 as the headline, with quadratic weighted kappa and an ordinal error rate beside it: confusing LOW with HIGH is not the same error as confusing LOW with MEDIUM.

- **Claims are machine-checked**
  Every scope claim is stored as the sentence it would appear as in the report, paired with an assertion run against the corpus. A failing assertion blocks table generation. There is no override flag.

REPORTED AS

- **Mean ± 95% CI**
  `bootstrap, 10,000 resamples`
- **Paired t-test**
  `α = 0.05, 30 paired seeds`
- **Effect size**
  `Cohen's d, always alongside p`
- **Fold count beside the mean**
  `4 points is not 30`
- **Seeds fixed, CSVs committed**
  `generated, never transcribed`

AND WHERE IT IS CIRCULAR

Labels come from detector counts, and three baselines also consume them — scoring those on auto-generated labels scores them against their own input. So the test split is human-verified; training data stays auto-labelled.

The ablation, reported as measured

4 folds × 5 seeds = 20 runs per configuration. Nothing below is a single-seed number.

CFG	VARIANT	MACRO-F1	[95% CI]	ACC	SEED SPREAD
A	CNN only	0.3596	[0.3144, 0.4051]	0.5318	0.1604
B	ViT only	0.4059	[0.3607, 0.4491]	0.5624	0.1931
C	**Concatenation**	**0.4405**	**[0.3724, 0.5124]**	**0.5996**	**0.1915**
D	One-way cross-attention	0.2787	[0.2094, 0.3475]	0.4281	0.1247
E	Bidirectional, no gate	0.4123	[0.3333, 0.4870]	0.5777	0.2090
F	Gated + temporal attention	0.3900	[0.3077, 0.4675]	0.5429	0.2136
G	Full model	0.3950	[0.3226, 0.4642]	0.5513	0.1984
H	Linear probe	0.4155	[0.3797, 0.4494]	0.5914	0.1649

COUNT-ONLY BASELINES, SAME FOLDS

gru counts	**0.5663**
naive last value	0.5571
lstm counts	0.5405
xgboost counts	0.4691

WHAT THIS SAYS, HONESTLY

The best vision configuration is C at 0.4405. A count-only sequence model reaches 0.5663 on the same folds and seeds. On this corpus the appearance features are not yet earning their place, so the vision claim is reported as open — not as established, and not dropped.

The control half already works

30 paired seeds against the swept, best-configuration Webster baseline.

19.35%	11.33 s	14.05 s	-1.34	p < 0.0001
lower mean waiting time	PPO, forecast-free arm	Webster, best config	Cohen's d	paired t-test

n = 30 paired seeds, saturated demand regime. This arm carries no forecast, so it measures the value of learned phase selection alone — the contribution of the forecast is a separate experiment and is not claimed here.

FULL MODEL vs LINEAR PROBE — PRE-REGISTERED

Cannot Distinguish

Mean -0.0204, 95% CI [-0.1544, 0.0657], detectable 0.2115 at 4 folds. Per fold 0:+0.0237 1:-0.2163 2:+0.0796 3:+0.0313: the full model leads on three, and the M6 fold — structured traffic, 52.6% of HIGH — reverses the mean. On the three in-regime folds it leads by 0.0449. Four-fold is the registered headline; both are reported.

WHAT THE CORPUS DOES NOT YET CONTAIN

- **No fold-eligible camera is Indian**
  three Bangladeshi, one UK motorway
- **HIGH is scarce in three of four folds**
  0 / 20 / 9 / 1 test windows
- **Both are failing assertions**
  report-table generation is blocked, by design

Part B campus footage is the work that clears them. Until it does, no claim about the Indian domain is made.

One laptop, one 6 GB GPU, ₹0

Nothing is listed because it is conventional. Two entries below are the reason the project is feasible at all.

HARDWARE

- **Training workstation**
  Laptop, 8-core x86-64 CPU, 16 GB RAM
- **GPU**
  NVIDIA RTX 4050 Laptop, 6 GB VRAM, compute capability 8.9
- **Storage**
  512 GB SSD, of which roughly 60 GB for video, cache and weights
- **Edge node**
  Team laptop acting as the edge device (ADR-003)
- **Camera**
  Existing municipal or campus CCTV stream, 720p or better
- **Network**
  Wi-Fi or Ethernet LAN; MQTT broker within 200 ms

SOFTWARE

- **Operating system and language**
  Windows 11 and Ubuntu 22.04 (both supported) · Python 3.12+
- **Deep learning**
  PyTorch 2.13.0 with CUDA 12.6 · torchvision 0.28.0
- **Reinforcement learning and simulation**
  Stable-Baselines3 2.9.0 · Gymnasium 1.3.0
- **Serving and deployment**
  ONNX 1.22.0 and ONNX Runtime 1.29.0
- **Data, storage and analysis**
  SQLite (embedded; no separate database server)
- **Engineering tooling**
  Git with Git LFS for model weights

Training is local, not cloud Caching the frozen backbones collapses the full A–H ablation from ≈12.3 h on CPU to ≈1.5 h; an uncached step does not fit in 6 GB.

SQLite, not a database server The prototype stores a bounded history of counts and decisions; a separate server would be infrastructure without a purpose.

Seven deliverables, each with a stated target

Targets are written down in advance so a miss is visible as a miss rather than reframed afterwards.

DELIVERABLE	MEASURABLE TARGET	STATUS
Curated dataset	Part A curated + 1,500-3,000 campus frames, per-camera provenance record	Part A done, Part B pending
Fine-tuned detector	≥ 0.85 mAP@0.5; cross-camera reported separately	Done
MFSTNet model	Weights + ONNX export; gate exposed as an artefact	Done
Ablation & evaluation	All 8 configs × 4 folds × 5 seeds, bootstrap CIs from a committed script	Done; underpowered
PPO controller	Significant reduction vs best Webster over 30 paired seeds — met at 19.35%	Done
Working prototype	Edge node, MQTT contract, control server, two-panel dashboard	Dashboard pending
Reproducibility package	Pinned deps, fixed seeds, committed raw CSVs, public release	Done

ACADEMIC OUTCOME

A conference submission whose contribution is the camera-only forecast-to-control path, the gate reported as an artefact, and a leave-one-camera-out evaluation with the linear-probe floor beside it. Under that framing an inconclusive vision result, reported with its statistical power, is still a contribution.

Where this stands, and what is next

WHAT WORKS

The whole pipeline is implemented end to end — corpus, detector, feature cache, MFSTNet with all eight configurations, leave-one-camera-out evaluation, SUMO environment, PPO arms, benchmark harness, ONNX export, control server and edge node with both fallbacks. The control result is real: 19.35% lower mean waiting time than the best-configured Webster baseline over 30 paired seeds.

WHAT DOES NOT, YET

The vision claim is open. Configuration C reaches 0.4405 macro-F1 against 0.5663 for a count-only sequence model, and the full model cannot be separated from the linear probe at four folds. No fold-eligible camera is Indian, and the dashboard is not built. All four are reported, not deferred.

1. **Expand the corpus with Indian campus footage** so the domain claim can be made honestly and the claims checker goes green
2. **Raise the number of evaluable cameras** so the architecture comparison has the power to resolve rather than returning “cannot distinguish”
3. **Complete the forecast-carrying control arm** so the value of the forecast is measured rather than assumed
4. **Build the descoped two-panel dashboard** benchmark numbers and live numbers visually separated, never merged

Expected significance: a low-cost, privacy-preserving adaptive signal control path for unstructured urban traffic, delivered with the evidence needed to judge whether each of its parts earns its place.

References (IEEE style)

Cited at first use in the synopsis report; numbering matches the report exactly.

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